Questions

Q1.

An atom has a central nucleus containing neutrons and protons.

Electrons orbit the nucleus.

One isotope of carbon is carbon-14.

$$^{14}_{6}C$$

(i) State the number of protons in one atom of carbon-14.

(1)

number of protons = ...

(ii) State the number of neutrons in one atom of carbon-14.

(1)

number of neutrons = ...

(iii) Figure 6 shows a graph for the decay of the radioactive isotope carbon-14.

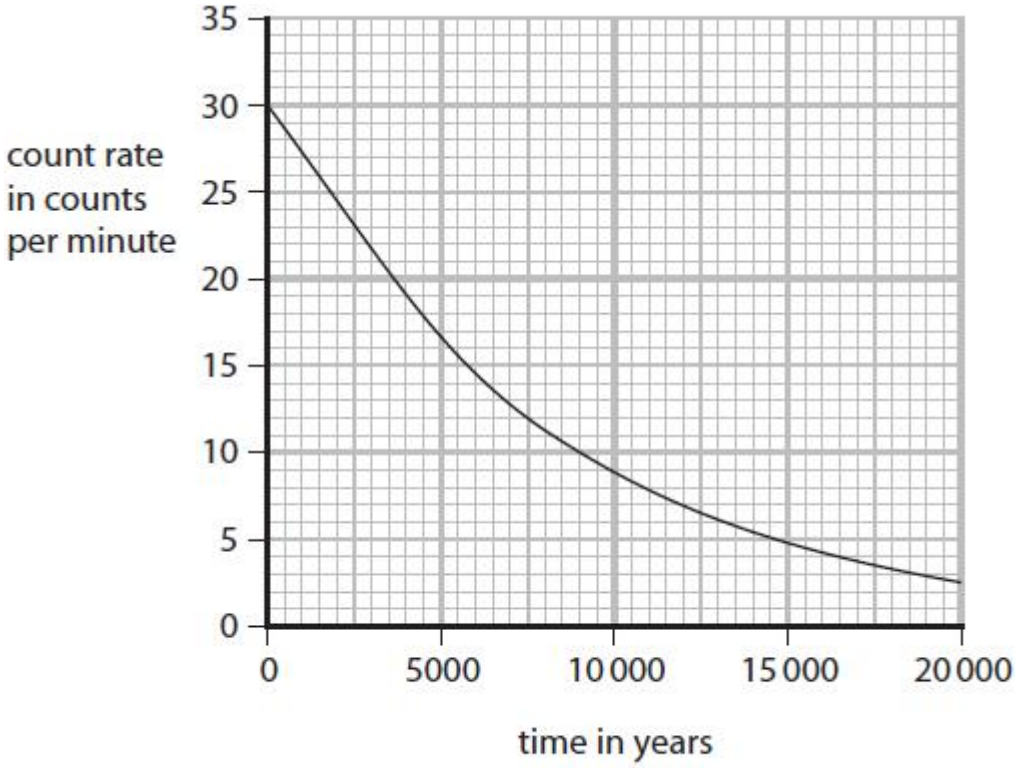

Figure 6

Use the graph to estimate the half-life of carbon-14.

(2)

half-life = ... years

(Total for question = 4 marks)

Q2.

Figure 17 is a diagram of a nuclear reactor.

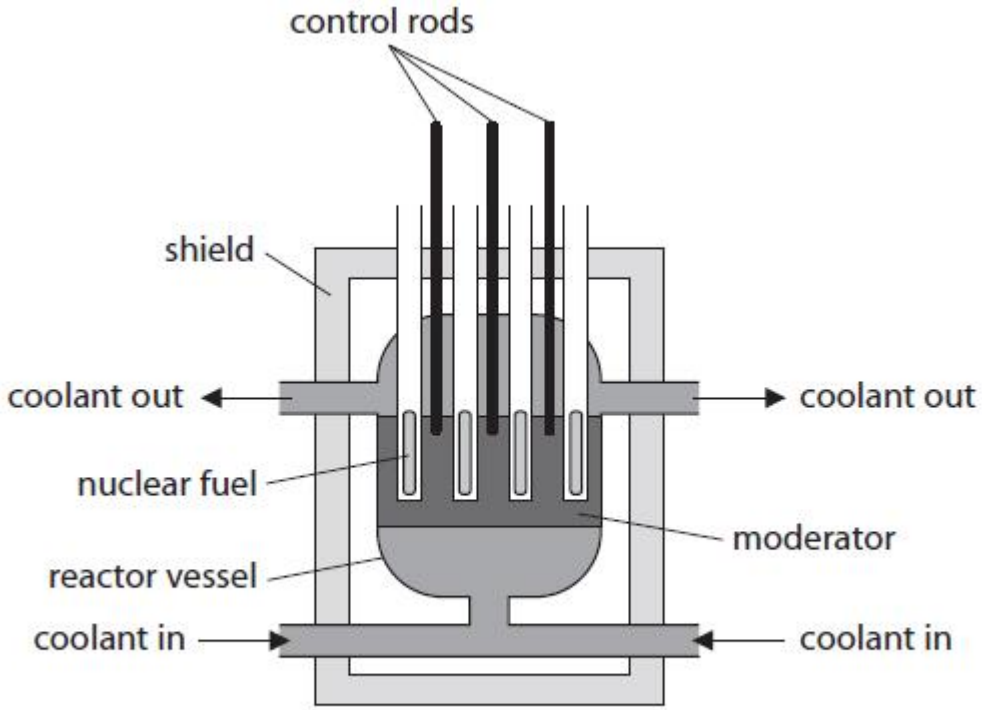

Figure 17

(i) Explain how pushing the control rods further into the reactor slows down the nuclear chain reaction.

(2)

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(ii) The moderator in a nuclear reactor slows down the neutrons so that the neutrons are more likely to start other fission reactions.

In a nuclear reactor,

- the average speed of the fast neutrons is 3.0 × 107 m/s
- the average speed of the slow neutrons is 4.0 × 103 m/s

Calculate the average speed of the slow neutrons as a percentage of the average speed of the fast neutrons.

(2)

..... %

(iii) The nuclear reaction is the first stage in the process of generating electricity.

Describe how energy is transferred from the nuclear reaction to the next stage in the process.

(2)

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(Total for question = 6 marks)

Q3.

Figure 17 shows a Geiger-Müller (GM) tube used for measuring radioactivity.

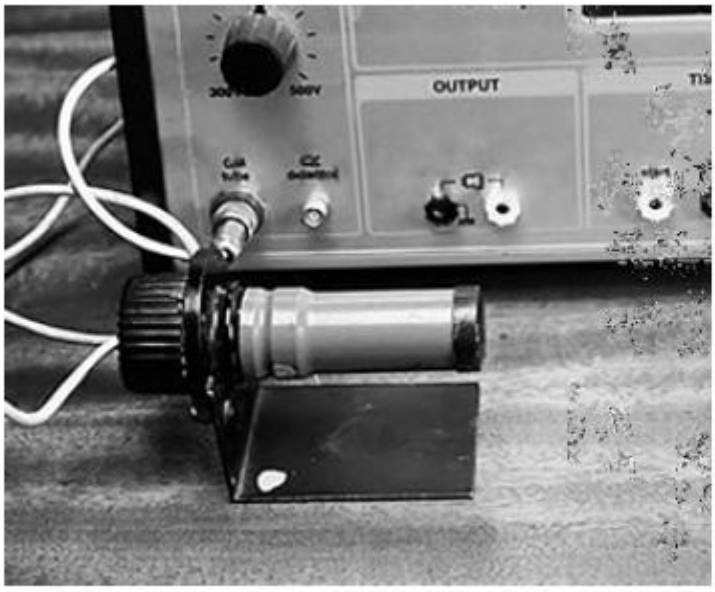

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Figure 17

* A radioactive rock is placed near to the front of a Geiger-Müller (GM) tube.

A radioactivity count-rate is first made in air.

The count-rate is measured again with each of three different absorbers between the rock and the GM tube.

Figure 19 shows the count-rates measured.

absorber	count-rate in counts per minute
3 cm of air	1272
thin sheet of paper	931
3 mm thick sheet of aluminium	328
2 cm thick sheet of lead	21

Figure 19

A scientist has an idea that the rock emits three different types of radiation.

Explain how the data in this table supports the scientist's idea.

(6)

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(Total for question = 6 marks)

Q4.

Figure 8 shows a helium nucleus.

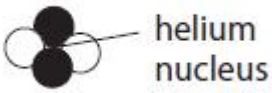

Figure 8

The energy released per kilogram of fuel in a fusion reaction is 845 000 GJ.

The energy released per kilogram of fuel in burning oil is 0.0394 GJ.

(i) Calculate the ratio of the energy released in fusion compared with the energy released in burning oil.
Use the equation

$$\text{ratio} = \frac{\text{energy released from fusion}}{\text{energy released by burning oil}}$$

(2)

ratio = ...

(ii) State **two** advantages of using a fusion reactor rather than burning oil in a power station.

(2)

1 .....

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2 .....

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(iii) State **two** of the difficulties that need to be overcome to produce a fusion reactor.

(2)

1 .....

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2 .....

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(Total for question = 6 marks)

Q5.

The fuel in a nuclear power station is an isotope of uranium.

Describe how the thermal energy produced by the nuclear reaction is used to produce electricity. You may draw a diagram to help with your answer.

(2)

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Q6.

Describe how the emissions from radioactive substances can be dangerous to living things.

(2)

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Q7.

***** Medical physicists have developed endoscopes and many other devices to help doctors diagnose medical problems.

Compare the use of electromagnetic radiation in endoscopes and in one other diagnostic device.

(6)

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Q8.

Sulfur-35 is a radioactive isotope of sulfur.
Figure 11 represents a nucleus of sulfur-35.

$$^{35}_{16}S$$

Figure 11

Draw one line from each type of particle to the number of that type of particle in a nucleus of sulfur-35.

(3)

type of particle	number of particles
proton	35
neutron	16
nucleon	51
	19

(Total for question = 3 marks)

Q9.

Beryllium-9 is a stable isotope of beryllium.
(a) (i) State the meaning of the term **stable.**

(1)

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(ii) Beryllium-9 has an atomic number of 4 and a mass number of 9.
A nucleus of this isotope can be described using this symbol.

$$^{9}_{4}Be$$

Complete the sentence by putting a cross (☒) in the box next to your answer.
The number of neutrons in this nucleus is

(1)

A 4

B 5

C 9

D 13

(iii) Which one of these symbols describes the nucleus of a different isotope of beryllium?

Put a cross (☒) in the box next to your answer.

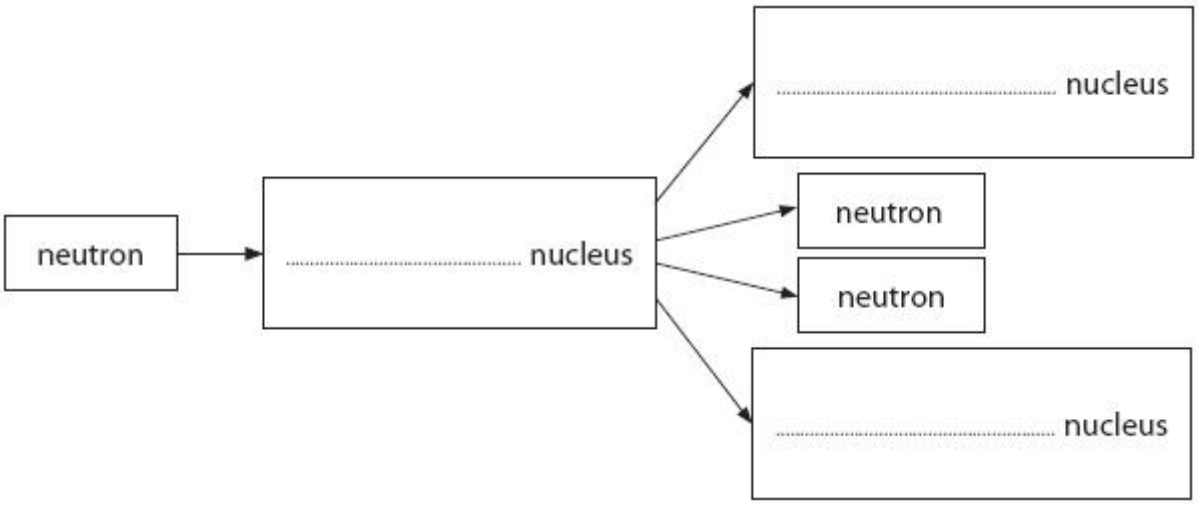
(1)

$^{8}_{4}Be$	$^{8}_{5}Be$	$^{9}_{4}B$	$^{9}_{5}Be$
☐ **A**	☐ **B**	☐ **C**	☐ **D**

(b) A beryllium-9 nucleus absorbs a neutron.
After a short time the new nucleus splits into two neutrons and two alpha particles.

(i) Complete the flow chart for this reaction.

(2)



(ii) Compare this nuclear reaction with the fission of a uranium nucleus.

(3)

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(iii) A fission reaction can be the start of a chain reaction.

Describe what needs to happen next to produce a chain reaction.

(2)

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(Total for Question = 10 marks)

Q10.

A scientist investigates a sample of an unknown radioactive material.

The scientist finds that the sample does **not** emit beta radiation.

The scientist sets up the experiment shown in Figure 18 and uses different absorbers to investigate alpha and gamma radiation.

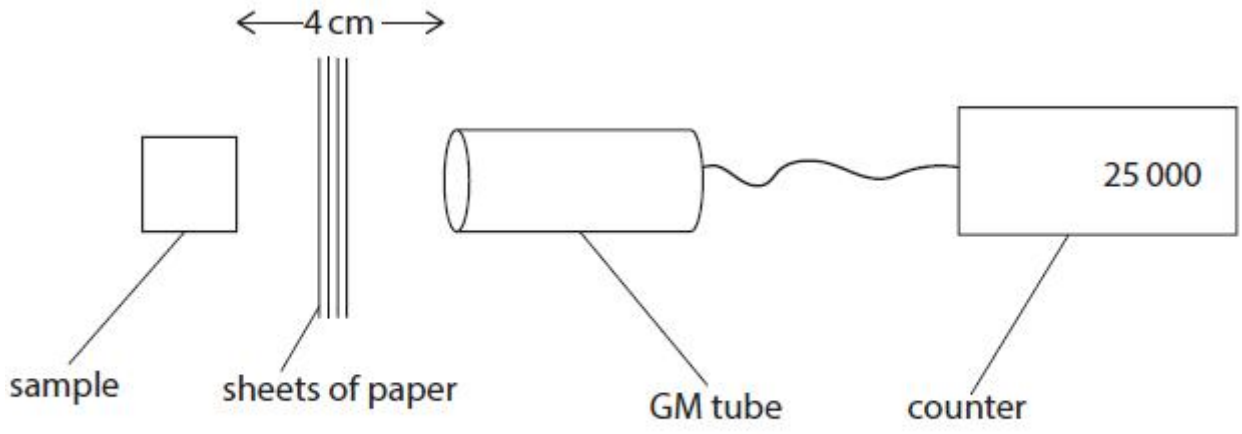

Figure 18

The results are shown in Figure 19.

The count rates are rounded to two significant figures.

The background radiation was 18 counts per minute (cpm).

absorber	count rate / cpm
none	33 000
4 sheets of paper	25 000
3 cm of lead	4 000

Figure 19

Discuss whether the sample emits alpha or gamma radiation or both.

Your answer should refer to

- the data shown in Figure 19
- how the data gives evidence for particular radiations.

(6)

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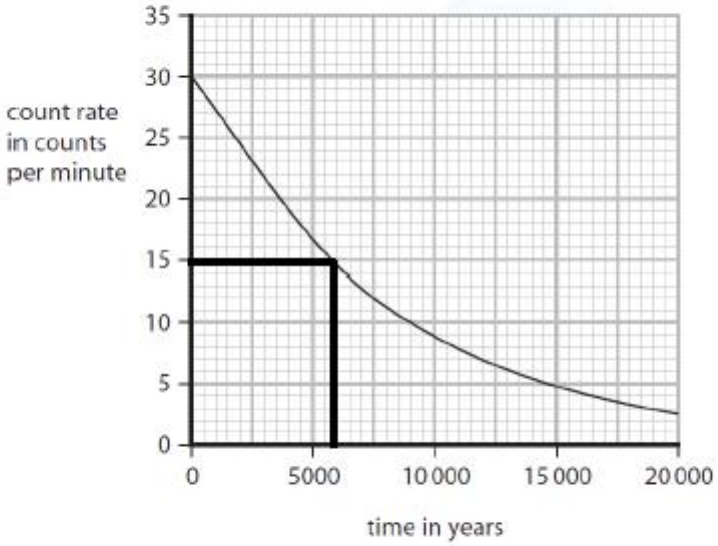
(Total for question = 6 marks)

Mark Scheme

Q1.

Question Number	Answer	Additional guidance	Mark
(i)	6 / six		(1) AO1

Question Number	Answer	Additional guidance	Mark
ii	8 / eight		(1) AO2

Question Number	Answer	Additional guidance	Mark
(iii)	indication of horizontal line between 14 and 16 and / or vertical line between 5250 and 6250 (1)  value between 5250 (years) and 6250 (years) inclusive (1)	accept alternative indications e.g. cross on curve accept any halving pairs e.g. going between 20 cpm and 10 cpm award full marks for the correct answer with no working	(2) AO3

Q2.

	Answer	Additional guidance	Mark
(i)	an explanation linking:	ignore slow down the nuclear chain reaction	(2) AO1
	(control rods) absorb neutrons (1)	accept (control rods) block neutrons accept neutrons can't pass through (control rods)	
	(so) fewer (neutrons) available for chain reaction (1)	fewer fission(s) (reactions)	

	Answer	Additional guidance	Mark
(ii)	$\frac{4(.0 \times 10^3)\ (\times 100)}{3(.0 \times 10^7)}$ (1)		(2) AO2
	1.3×10^{-2} (%) (1)	0. 013 (%) allow 0.01 (%) power of ten error scores 1 mark maximum award full marks for the correct answer without working	

	Answer	Additional guidance	Mark
(iii)	A description to include: (from) kinetic energy (of fission fragments) (1) (transferred to) thermal energy (of coolant) (1)	accept references to energy stores accept energy in nuclear store accept nuclear energy / gamma radiation energy / binding energy / mass (to) thermal store (in coolant) accept heat for thermal allow steam transfers thermal energy/heat from reactor to kinetic energy of turbine for 2 marks	**(2)** **AO1**

Q3.

Question Number	Answer	Mark
	Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive, and candidates are not required to include all the material which is indicated as relevant. Additional content included in the response must be scientific and relevant. **AO3 Strand 2a and 2b (6 marks)** • shows some idea that the data can support arguments about alpha, beta and gamma radiation being present • argues that there is some evidence that alpha might be emitted (count rate going down with paper interposed) • argues that there is a lot of evidence that beta particles are emitted (count rate goes down a lot when the aluminium is inserted) • argues that there might be some gamma getting through (lead stopping everything apart from gamma) OR that with the lead present the count rate has gone down to a level consistent with background, so no gamma was present a level 3 answer will use data effectively	**(6)** AO 1 1

Level	Mark	Descriptor
	0	• No rewardable material.
Level 1	1-2	• Deconstructs scientific information but understanding and connections are flawed. An unbalanced or incomplete argument that provides limited synthesis of understanding. • Judgements are supported by limited evidence. (AO3)
Level 2	3-4	• Deconstructs scientific information and provides some logical connections between scientific concepts. An imbalanced argument that synthesises mostly relevant understanding, but not entirely coherently. • Judgements are supported by evidence occasionally. (AO3)
Level 3	5-6	• Deconstructs scientific information and provide logical connections between scientific concepts throughout. A balanced, well-developed argument that synthesises relevant understanding coherently. • Judgements are supported by evidence throughout. (AO3)

Q4.

Question Number	Answer	Additional guidance	Mark
(i)	substitution (1) $\frac{845\ 000}{0.0394}$ evaluation (1) 21 000 000	answers that round to 21 000 000 $2.1(45) \times 10^7$ etc. award full marks for the correct answer without working	**(2)** AO 2 1

Question Number	Answer	Additional guidance	Mark
(ii)	any two from: • fusion power gives (many) more times the energy output (for the same mass used) • no greenhouse gases / CO_2 emissions (produced with the fusion alternative) • does not lead to global warming • no (radioactive) waste • does not deplete / use up a finite resource (e.g. oil)	may quote numbers here accept no or less pollution / no or less harmful gases etc. sustainable reference oil is running out ignore references to costs	(2) AO 1 1

Question Number	Answer	Additional guidance	Mark
(iii)	any two from: • problem of containment (the fusion gases / isotopes at high temperatures) • (maintaining) high temperature • (maintaining) high pressure		(2) AO 2 1

Q5.

	Answer	Acceptable answers	Mark
	An explanation linking any two of the following points • heats/boils water (1) • to produce steam (1) • to run/turn/spin turbines (1) • to turn/power generators (1)	labelled diagram that indicates process (not just parts). heats boiler turns a coil in a magnet	**(2)**

Q6.

	Answer	Acceptable answers	Mark
	A description including **two** of: • Kill/damage cells(1) • affecting DNA (1) • (causing) mutation (1) • by ionisation (1) • make cell reproduce rapidly (1) • cause cancer (1) • (radiation) burns (1) • (radiation) sickness (1)		**(2)**

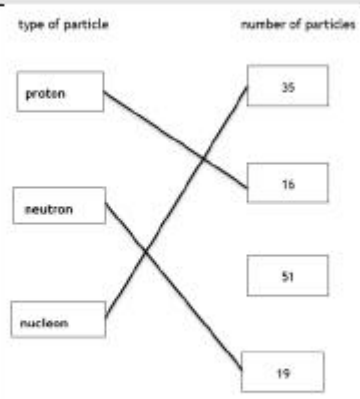
Q7.

		Indicative Content	Mark
QWC	*	A comparison of endoscopes with any one of the following devices: **Diagnostic devices** • CAT scanners • Fluoroscopes • Thermal imagers / IR thermometers • Pulse oximeters • PET scanners • X-ray machines • Gamma cameras**Link to electromagnetic radiation** • Endoscopes use TIR of light in optical fibres • CAT scanners X-rays and computer to generate 3D images • Fluoroscopes use X- rays and a video camera • Thermal imagers use infrared emitted by a body • IR / red LEDs used to measure oxygen levels	**(6)**

		• PET scanners detect radiation emitted by electronpositron annihilation • Gamma cameras detect gamma rays from radioactive sources**Other factors for comparison** • Safety • Ease of use • Frequency / wave length • Intensity • Penetration • Ionising / non-ionising	
Level	**0**	No rewardable content	
1	**1 - 2**	• a limited comparison between an endoscope and one device e.g. endoscopes use light and CAT scanners detect broken bones • the answer communicates ideas using simple language and uses limited scientific terminology • spelling, punctuation and grammar are used with limited accuracy	
2	**3 - 4**	• a simple comparison between an endoscope and one device, linking them to the electromagnetic radiation used for both and a detail of use for one of them e.g. endoscopes use visible light to examine internal organs and CAT scans use X-rays • the answer communicates ideas showing some evidence of clarity and organisation and uses scientific terminology appropriately • spelling, punctuation and grammar are used with some accuracy	
3	**5 - 6**	• a detailed comparison between an endoscope and one device, linking them to the electromagnetic radiation used for both and a detail of use for both of them e.g. endoscopes use visible light which is passed down optical fibres by TIR to examine internal organs. Fluoroscopes use X-rays and a video camera to show positioning of stents in arteries. • the answer communicates ideas clearly and coherently uses a range of scientific terminology accurately	

		• spelling, punctuation and grammar are used with few errors

Q8.

	Answer	Additional guidance	Mark
		1 mark for each correct line more than one line from a box in the left column ("type of particle") box loses the mark for the box	(3) AO2

Q9.

	Answer	Acceptable answers	Mark
(a)(i)	does not emit (ionising) radiation / no (radioactive) decay	it is not radioactive	**(1)**
(a)(ii)	$^{8}_{4}Be$ **B** 5		**(1)**
(a)(iii)	[redacted] $^{8}_{4}Be$ A		**(1)**
(b)(i)	heli um beryllium (1) (1) he lium	daughter in right hand boxes daughter	**(2)**
(b)(ii)	a comparison which describes any **three** of the following: similarities: • produce (more) neutrons (1) • produce 'daughter' (nuclei) (1)	different elements / smaller nuclei for daughters do not accept split an atom neutron is absorbed	**(3)**

	• release energy (1) • split a (bigger) nucleus (1) • (triggered by) a neutron coming in (1) • nucleus becomes unstable (before splitting) (1)differences: • uranium daughters are different from each other/ beryllium daughters are the same (1) • uranium daughters are heavier than beryllium daughters (1)		
(b)(iii)	a description including: neutron(s) (from first fission) (1) (go on to) cause another fission (1)	collide with another nucleus /atom	**(2)**

Q10.